Polynomial Regression Assignment

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1 Introduction

This assignment applies polynomial regression to two prediction problems. Var1 predicts Net Power Score from six input variables, while Var2 predicts Thermal Anomaly Score from three spatial variables.

Polynomial degree and Elastic Net parameters are selected using five-fold cross-validation, after which the selected models are trained on the complete training datasets and used to predict the hidden test sets.

2 Methodology

2.1 Polynomial Regression

Polynomial features and interaction terms are generated using `PolynomialFeatures`. Degrees 1–10 are tested for Var1 and degrees 1–20 for Var2.

Since polynomial expansion can produce features with different scales, `StandardScaler` is applied before regression. No min-max normalization is performed because the original input variables are already on a comparable scale.

2.2 Elastic Net and Model Selection

Elastic Net combines L1 and L2 regularization. The regularization strength is controlled by α, while β denotes the L1 ratio.

$$\text{Loss} + \alpha \left[\beta \|w\|_1 + (1-\beta)\|w\|_2^2\right].$$

Five-fold shuffled cross-validation is used to select the polynomial degree and Elastic Net hyperparameters. The configuration with the lowest mean validation MSE is selected.

2.3 Evaluation

Mean Squared Error is used for model selection:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2.$$

Training R^2 is also recorded. Since the test targets are hidden, test-set MSE and R^2 cannot be calculated locally.

3 Results

3.1 Var1 – Net Power Score

The best configuration for Var1 is degree 5.

Table 1: Var1 selected model

Degree	5
α	0.0083268
β	0.95
CV MSE	0.333396
Training MSE	0.227370
Training R^2	0.976938

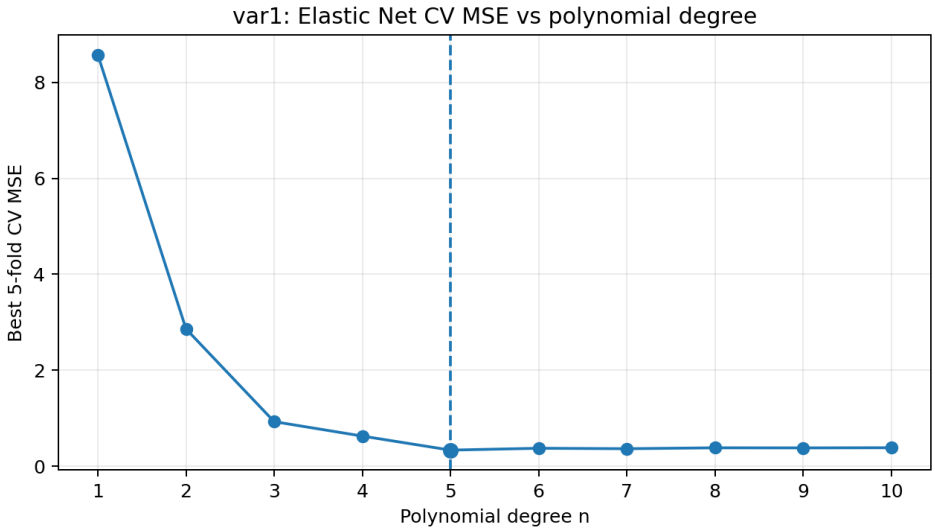


Figure 1: Var1 CV MSE versus polynomial degree.

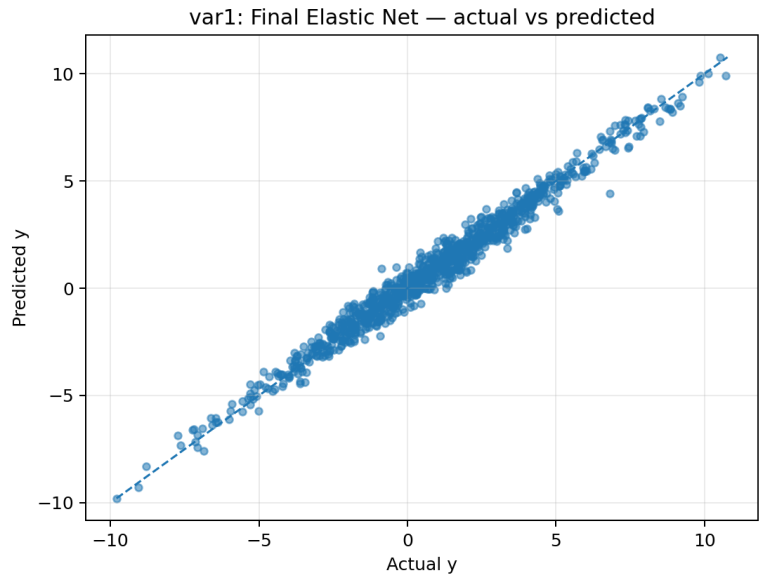


Figure 2: Var1 actual versus predicted values.

3.2 Var2 – Thermal Anomaly Score

The best configuration for Var2 is degree 12.

Table 2: Var2 selected model

Degree	12
α	0.0057735
β	0.05
CV MSE	0.253945
Training MSE	0.189553
Training R^2	0.996291

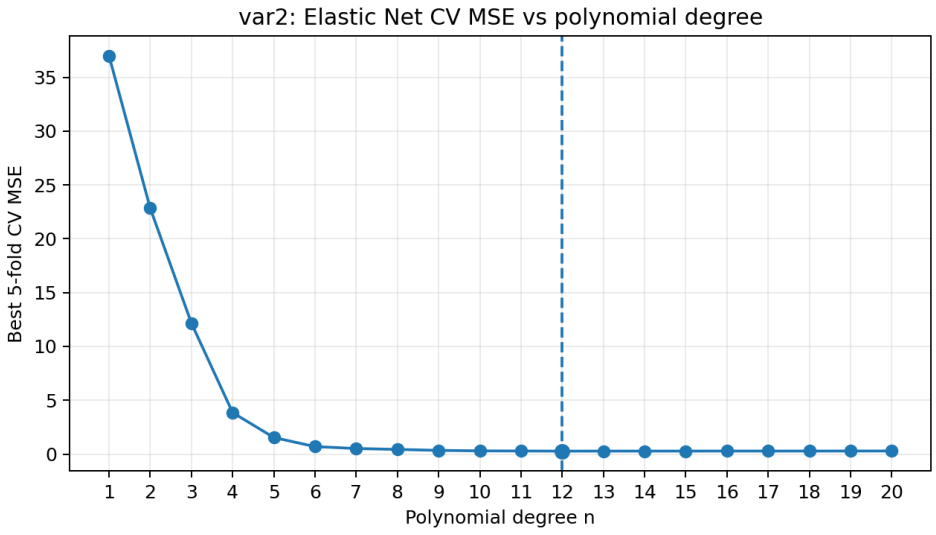


Figure 3: Var2 CV MSE versus polynomial degree.

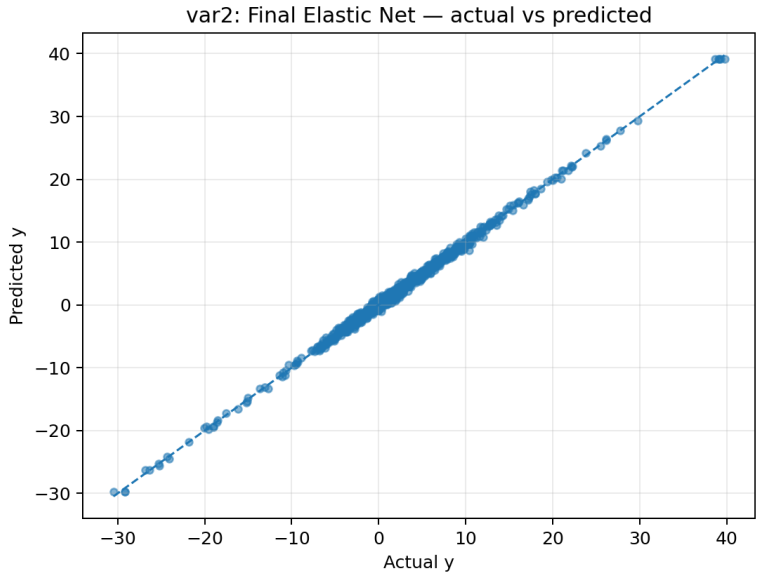


Figure 4: Var2 actual versus predicted values.

4 Final Model Selection

Table 3: Final selected models

Problem	Degree	α	β	CV MSE
Var1	5	0.0083268	0.95	0.333396
Var2	12	0.0057735	0.05	0.253945

The selected models were retrained using all 1,000 training observations and used to generate predictions for the hidden test sets.

5 Conclusion

Elastic Net polynomial regression successfully provides regularized models for both problems. Cross-validation selected degree 5 for Var1 and degree 12 for Var2. The final models were then trained on the complete training data and used to generate the required test predictions.